Summary
This section develops the understanding of pressure in gases and liquids. Pressure in gases builds on the particle model, and in liquids the increase in pressure with depth is explained as the weight of a column of liquid acting on a unit area.

Underlying knowledge and understanding
Learners should be aware of the change in pressure in the atmosphere and in liquids with height (qualitative relationship only). They should have an understanding of floating and sinking and the effect of upthrust. Learners should know that pressure is measured by a ratio of force over area which is acting at a normal to the surface.

Common misconceptions
Learners commonly have misconceptions about floating and sinking, based on the premise that light or small objects float and heavy or large objects sink. They often misunderstand the role of pressure difference and suction e.g. the collapsing can experiment and the forcing of air into the lungs during inhalation.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM1.3i ☑	apply: for a given mass of gas at a constant temperature pressure (Pa) × volume (m^3) = constant	M1a, M3b, M3c, M3d
PM1.3ii ☑	**apply:** **pressure due to a column of liquid (Pa) = height of column (m) × density of liquid (kg/m^3) × gravitational field strength (N/kg)**	**M1a, M1c, M3b, M3c, M3d**

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P1.3a	explain how the motion of the molecules in a gas is related both to its temperature and its pressure	application to closed systems only	M1c, M4a, M5b	WS1.1b, WS1.2a, WS1.2e, WS1.3e, WS1.4a, WS2a	Demonstration of the difference in pressure in an inflated balloon that has been heated and frozen. (PAG P1) Building manometers and using them to show pressure changes in heated/cooled volumes of gas. (PAG P1)

P1.3b	explain the relationship between the temperature of a gas and its pressure at constant volume (qualitative only)		M1c, M5b	WS1.1b, WS1.2a, WS1.2e, WS1.3e, WS1.4a, WS2a	Demonstration of the exploding can experiment. Building of Alka-Seltzer rockets with film canisters.
P1.3c ☑	recall that gases can be compressed or expanded by pressure changes and that the pressure produces a net force at right angles to any surface		M4a, M5b	WS1.1b, WS1.2a, WS1.2e, WS1.3e, WS1.4a, WS2a	Compressing syringes containing sand, water and air. (PAG P1) Demonstration of the collapsing can experiment. Demonstration of the Cartesian diver experiment.
P1.3d ☑	explain how increasing the volume in which a gas is contained, at constant temperature can lead to a decrease in pressure	behaviour regarding particle velocity and collisions	M1c, M4a, M5b	WS1.1b, WS1.2a, WS1.2e, WS1.3e, WS1.4a	Demonstration of the behaviour of marshmallows in a vacuum.
P1.3e ☑	**explain how doing work on a gas can increase its temperature**	**examples such as a bicycle pump**		WS1.1b, WS1.2a	Demonstration of heat production in a bicycle inner tube as it is pumped up.
P1.3f ☑	describe a simple model of the Earth's atmosphere and of atmospheric pressure	an assumption of uniform density; knowledge of layers is not expected	M5b		
P1.3g ☑	explain why atmospheric pressure varies with height above the surface of the planet				

P1.3h ☑	**describe the factors which influence floating and sinking**				
P1.3i ☑	**explain why pressure in a liquid varies with depth and density and how this leads to an upwards force on a partially submerged object**			WS1.1b, WS1.2a, WS1.3a, WS2a	Discussion of buoyancy of a ping pong ball in water.
P1.3j ☑	**calculate the differences in pressure at different depths in a liquid**	**knowledge that *g* is the strength of the gravitational field and has a value of 10 N/kg near the Earth's surface**	M1c, M3c	WS1.1b, WS1.2a	Demonstration of differences in water pressure using a pressure can with holes.